

CSE437 Data Science - Project Report

Cover

- **Project title:** Predicting Building Energy Use Intensity in Seattle Using Machine Learning •
- **Course:** CSE437 Data Science
- **Section:** 05
- **Semester:** Summer 2026
- **Group:** 09
- **Group members:**
 - Tashrique Tanvir - ID [23301389]
- **GitHub repository:** <https://github.com/tashriquetanvir-commits/cse437-building-energy-G9.git>
- **Date:** September 3, 2026

Summary

This project predicts **Site Energy Use Intensity (SiteEUI)**, measured in kBtu per square foot, from non-leaking physical and operational characteristics in the City of Seattle's *Building Energy Benchmarking Data, 2015-Present*. The downloaded data contained 38,309 building-year records and 46 columns; after retaining compliant records, removing missing targets, and excluding 34 extreme SiteEUI values above 800, 35,114 records remained. Ridge Regression and Random Forest Regressor were compared with an 80/20 building-grouped holdout and five-fold grouped cross-validation. The target was modelled as $\log_{10}(\text{SiteEUI})$ to reduce severe right-skew, and log-scale R^2 was declared the primary metric. Random Forest achieved the best test log- R^2 (0.420) and the lowest raw-scale MAE (20.49 kBtu/sf), while Ridge achieved the best raw-scale R^2 (0.273). The central finding is that building use type is the clearest available signal: supermarkets, laboratories, hospitals, recreation facilities, and data-centre-related uses have the largest positive effects. However, high-intensity specialist buildings remain the main failure cases, showing that physical metadata cannot fully represent occupancy, equipment loads, or operating schedules.

1. Problem and Dataset

1.1 Problem statement

Energy use intensity makes buildings of different sizes comparable by expressing annual site energy per square foot. Predicting SiteEUI from building characteristics can support early screening of inefficient buildings, prioritisation of audits, and more targeted energy-management decisions when complete consumption data are not yet available. The task is supervised regression: estimate a continuous SiteEUI value from attributes known about the building and its uses, while excluding any column that directly contains energy consumption or is mathematically derived from the target.

1.2 Dataset

Source: City of Seattle Open Data, *Building Energy Benchmarking Data, 2015-Present*: <https://data.seattle.gov/d/teqw-tu6e>. **Provider:** Seattle Office of Sustainability and Environment. **Collection method:** annual owner reported benchmarking through ENERGY STAR Portfolio Manager for covered multifamily and non-residential properties. **Downloaded size:** 38,309 rows × 46 columns. **Period in the downloaded file:** 2015-2025. **Terms:** public-domain dataset as stated in the portal metadata.

1.3 Target variable

The target is SiteEUI(kBtu/sf), a continuous variable. In the final cleaned sample ($n = 35,114$), mean = 51.11, median = 36.70, standard deviation = 49.09, minimum = 0.10, and maximum = 787.20 kBtu/sf. The raw distribution is strongly right-skewed (Figure 1).

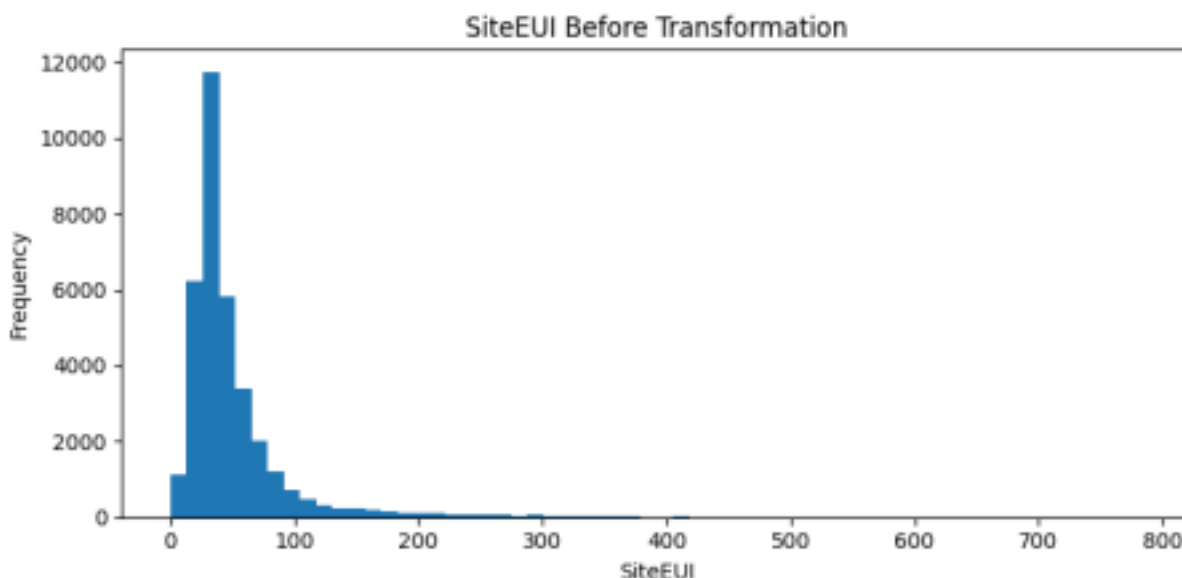


Figure 1: SiteEUI distribution before the log transformation.

1.4 Three questions

1. Which non-leaking building characteristics are most strongly associated with SiteEUI? 2. How accurately can Ridge Regression and Random Forest predict SiteEUI for previously unseen buildings? 3. Which model and feature representation performs best after preprocessing, dimensionality-reduction testing, and hyperparameter tuning?

2. Data Handling and Preprocessing

2.1 Data quality audit

- **Missing target:** 648 compliant records had no SiteEUI and could not be used for supervised learning.
- **Duplicates:** no exact duplicate rows and no duplicate OSEBuildingID-DataYear pairs were found. Repeated OSEBuildingID values are legitimate annual observations, not duplicates.
- **Inconsistent categories:** categorical values were stripped of surrounding whitespace and missing labels were handled inside the pipeline.
- **Impossible/extreme values:** SiteEUI was required to be positive. Thirty-four compliant observations above 800 kBtu/sf were excluded because they were isolated extremes that would dominate model fitting and evaluation.

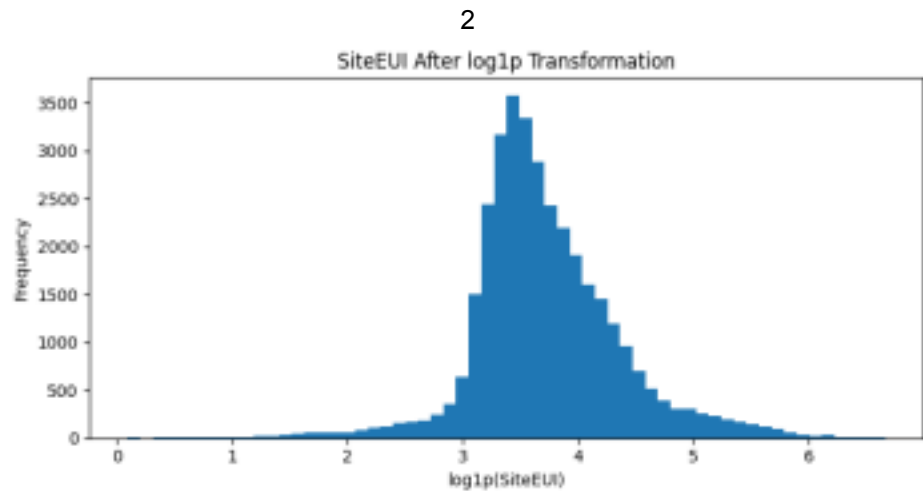
2.2 Missing values

Missing targets were dropped rather than imputed, because target imputation would fabricate the outcome. For predictors, numeric gaps were imputed with the training-fold median and categorical gaps with the training-fold most-frequent value. The second- and third-largest use fields had the most missingness; they were retained because absence also indicates that a property has no reported secondary or tertiary use. Every imputer was fitted only inside each training fold.

2.3 Outliers

IQR diagnostics flagged many legitimate high-use buildings, so the IQR rule was used for inspection rather than

automatic deletion. The modelling range was limited to 0.1-800 kBtu/sf, removing 34 cases above 800 from otherwise compliant, target-complete records. Values within this range were retained because specialist buildings such as hospitals and laboratories are genuinely energy intensive. The target was then log-transformed instead of winsorised.



2.4 Transformation and scaling

log1p(SiteEUI) was used as the modelling target. Numeric predictors were median-imputed and standardised for Ridge; Random Forest used median-imputed numeric values without scaling. Categorical predictors were most-frequent-imputed and one-hot encoded with unknown categories ignored. Transformations were contained in Pipeline/ColumnTransformer objects and fitted after the split, so information from validation and test build ings could not influence preprocessing.

Before splitting, the following leakage columns were removed in code: ENERGYSTARScore, SiteEUIWN, SiteEnergyUse, SiteEnergyUseWN, both SourceEUI fields, electricity/steam/natural-gas consumption totals, TotalGHGEmissions, and GHGEmissionsIntensity. Names, addresses, parcel numbers, and other identifiers were also excluded from predictors.

2.5 Before and after

Stage	Rows	Columns	Unique buildings	Change	
Raw download	38,309	46	3,871	Original combined annual file	
Compliant records	35,796	46	3,859	Removed non-compliant	
Target available	35,148	46	3,851	Dropped missing SiteEUI	reports
Final valid range	35,114	46	3,851	Removed 34 values above 800	
Model matrix	35,114	24	3,851	17 numeric + 7 categorical inputs	before encoding

3. Statistical Analysis

3.1 Descriptive statistics

The median SiteEUI (36.70) is substantially below the mean (51.11), confirming a long upper tail. The middle 50% lies between 27.50 and 56.20 kBtu/sf. The log transformation changes the target mean to 3.719 and standard deviation to 0.646, producing a much more compact distribution suitable for stable regression. There are 73 EPA property types, 14 neighbourhoods, and 7 broad building types. The dataset is longitudinal: 3,851 properties

contribute observations across up to eleven reporting years. This structure makes a normal row wise random split unsafe because it could place different years of the same building on both sides of the split.

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3.2 Relationships

Property use type has a strong practical relationship with SiteEUI. Among categories with at least 100 observations, median SiteEUI is highest for supermarkets/grocery stores (about 227), laboratories (210), and general hospitals (about 200). Offices have the lowest median among the fifteen displayed groups (about 47). These differences support including the reported property-use fields as categorical predictors.

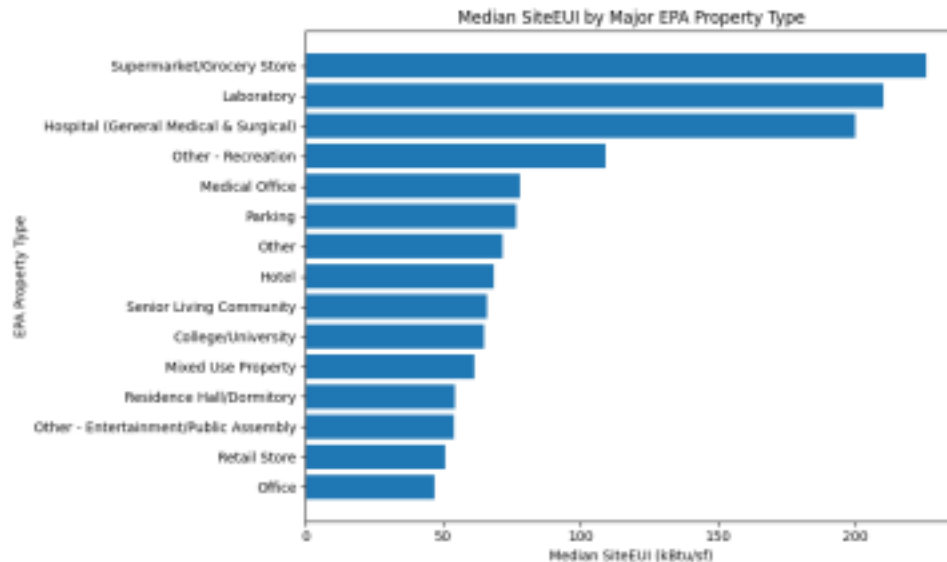


Figure 3: Median SiteEUI for EPA property types with at least 100 records, highest 15 shown.

3.3 What the data says so far

1. Raw SiteEUI is too right-skewed for an untransformed linear model.
2. Property use type is the clearest available explanatory signal.
3. Building size alone cannot explain intensity because SiteEUI is already normalised by area.
4. Annual records from the same building must remain in one split group.
5. Energy, emissions, SourceEUI, and ENERGY STAR fields must be removed to prevent target leakage.

4. Feature Engineering

4.1 Derived features

- $\text{BuildingAge} = \text{DataYear} - \text{YearBuilt}$, representing ageing and construction-era effects.
- $\text{ParkingFraction} = \text{PropertyGFAParking} / \text{PropertyGFATotal}$, separating parking-heavy sites from occupied floor area.
- $\text{LargestUseFraction} = \text{LargestPropertyUseTypeGFA} / \text{PropertyGFATotal}$, describing how dominant the main use is.
- $\log_{1p}(\text{SiteEUI})$ as the transformed training target; predictions were converted back with expm1 for MAE, RMSE, plots, and interpretation.

4.2 Dimensionality reduction

Truncated SVD was applied only to the preprocessed training matrix. Thirty components explained approximately 92% of cumulative variance (Figure 4), and roughly 25 components crossed 90%. SVD was evaluated as an alternative compact representation but was not kept in the final pipeline: retaining the full encoded space preserved direct Ridge coefficients and did not introduce a meaningful validation improvement.

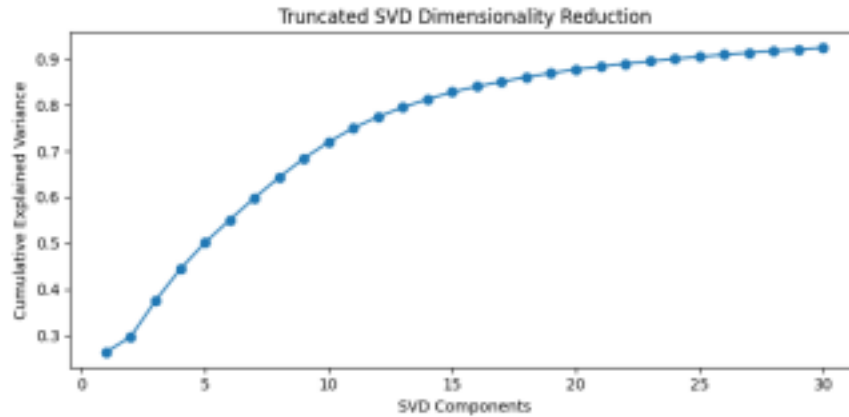


Figure 4: Cumulative explained variance from Truncated SVD.

4.3 Feature selection

Feature selection combined domain screening with model-based ranking. First, identifiers and all direct or indirect energy outcomes were removed. Second, categories were one-hot encoded. Finally, standardised Ridge coefficients were ranked by absolute magnitude; the report focuses on the top 15 coefficients. No hard coefficient threshold was used in the final model because small categorical effects can work jointly and Ridge regularisation already shrinks weak terms.

4.4 Final feature set

The final pre-encoding set contained 17 numeric and 7 categorical variables. Numeric inputs covered reporting year, building age, floor counts, total/building/parking floor areas, self-reported areas, use-type floor areas, area fractions, latitude, and longitude. Categorical inputs were building type, neighbourhood, council district, EPA property type, and the three largest reported property-use types. This set uses information plausibly available before the target is known and excludes high-cardinality identifiers.

5. Modeling and Validation

5.1 Validation strategy

An 80/20 GroupShuffleSplit with random_state=42 produced 28,059 training records from 3,080 buildings and 7,055 test records from 771 different buildings. Model selection used five-fold GroupKFold cross-validation on the training portion only, grouped by OSEBuildingID. The test set was evaluated once after preprocessing and tuning choices were fixed.

5.2 Baseline

A mean DummyRegressor on the log target was the trivial predictor. It achieved test log- $R^2 = -0.001$, raw-scale $R^2 = -0.052$, MAE = 27.07, and RMSE = 59.62 kBtu/sf after inverse transformation.

5.3 Model families

Ridge Regression is a regularised linear family suited to a large, correlated one-hot feature space. L2 regularisation stabilises coefficients and supports interpretation, but assumes additive linear effects on log-SiteEUI. **Random Forest Regressor** is a tree ensemble suited to non-linear relationships and interactions such as different size effects by use type. It carries fewer functional-form assumptions but can overfit rare property types and is less directly interpretable.

5.4 Metrics

Primary metric: R^2 on $\log_{10}(\text{SiteEUI})$, declared before test evaluation because this is the scale actually optimised and it limits domination by a few extreme buildings. Secondary metrics are raw-scale R^2 , MAE, and RMSE. MAE describes typical absolute error; RMSE gives extra weight to large misses.

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6. Hyperparameter Tuning

6.1 Search space

Model Hyperparameter Values searched

Ridge alpha 0.1, 1, 10, 30, 100
Random Forest n_estimators 100
Random Forest max_depth 10, 20, None
Random Forest min_samples_leaf 2, 4
Random Forest max_features sqrt, 0.5

6.2 Method

Ridge used an exhaustive five-candidate grid. Random Forest used a compact six-configuration grid selected from the combinations above to keep computation reasonable. Each candidate was assessed with five building-grouped folds and scored by log-scale R^2 . All preprocessing was re-fitted inside each fold.

6.3 Results

Model/configuration Mean grouped CV log- R^2

Ridge alpha = 0.1 0.308
Ridge alpha = 1 0.315
Ridge alpha = 10 0.327
Ridge alpha = 30 0.334
Ridge alpha = 100 0.335
RF depth = 10, leaf = 2, sqrt 0.319
RF depth = 20, leaf = 2, sqrt 0.376
RF depth = None, leaf = 2, sqrt 0.381
RF depth = 10, leaf = 4, features = 0.5 0.358
RF depth = 20, leaf = 4, features = 0.5 0.376
RF depth = None, leaf = 4, features = 0.5 0.373

Ridge improved gradually as regularisation increased. Random Forest underfit at depth 10; deeper trees improved validation performance, with unlimited depth, two observations per leaf, and square-root feature sampling performing best.

7. Results, Visualization and Error Analysis

7.1 Test set performance

Model Log- R^2 Raw R^2 MAE (kBtu/sf) RMSE (kBtu/sf)

Mean baseline -0.001 -0.052 27.07 59.62
Ridge (alpha = 100) 0.400 **0.273** 21.05 **49.57**
Random Forest (tuned) 0.420 0.253 **20.49** 50.26

Random Forest is selected on the declared primary metric and MAE. Ridge remains competitive and performs slightly better on raw-scale R^2 and RMSE, so the advantage is modest rather than decisive.

7.2 Visualization

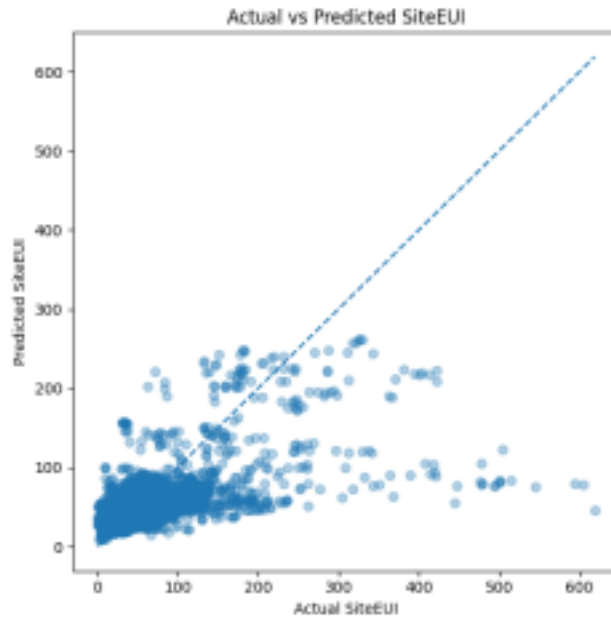


Figure 5: Actual versus predicted SiteEUI on the original scale.

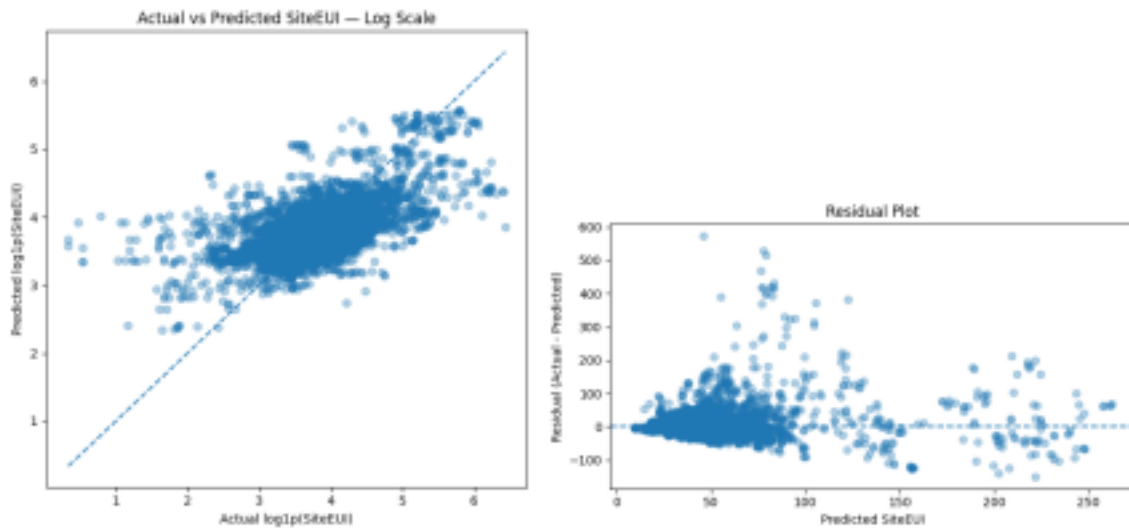


Figure 6:

Actual versus predicted values on the log scale (left) and raw-scale residuals against predictions (right).

Predictions follow the centre of the log-scale distribution more closely than the raw scale, confirming the benefit of the target transformation. The raw scatter and residual plots also show regression toward the mean: very high SiteEUI values are usually underpredicted, while the spread of residuals grows for more energy-intensive buildings.

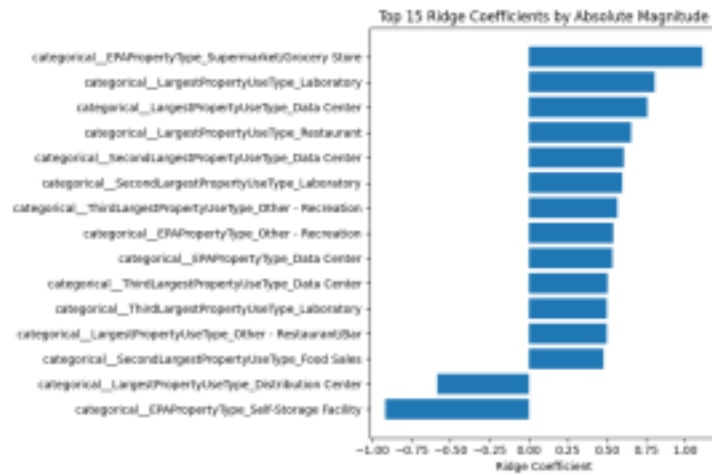


Figure 7: Top 15 Ridge coefficients ranked by absolute magnitude.

The coefficient ranking is dominated by property-use categories. Supermarket/grocery store, laboratory, data centre, restaurant, and recreation indicators push predicted log-SiteEUI upward. Self-storage and distribution centre indicators pull it downward. Because correlated use fields appear at several reporting levels, coefficient magnitudes should be interpreted as directional model effects rather than causal impacts.

7.3 Error analysis

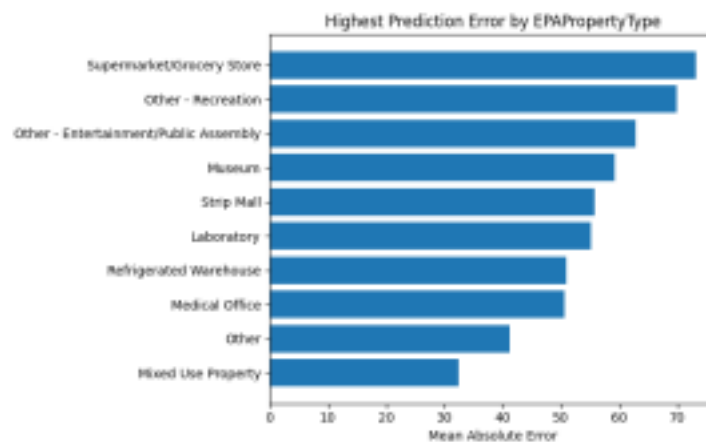


Figure 8: EPA property types with the highest mean absolute prediction error.

Errors are largest for supermarkets/grocery stores, recreation, entertainment/public assembly, museums, strip malls, laboratories, refrigerated warehouses, and medical offices. These groups have specialised loads, variable schedules, refrigeration, process equipment, or occupancy patterns that are not fully captured by floor area and use labels. **Example 1 - high-intensity supermarket.** A supermarket near 600 kBtu/sf is predicted below 100 kBtu/sf in the raw scatter, an error above 500. A category label identifies the broad use but not refrigeration age, operating hours, or equipment efficiency.

Example 2 - recreation/public assembly building. A record above 400 kBtu/sf is predicted near 100, an error of roughly 300. Event frequency, pools/ice systems, and intermittent occupancy can vary sharply between buildings with similar metadata.

7.4 Answers to the three questions

Q1 - Which characteristics matter most? Property-use categories are the clearest predictors. Median comparisons and Ridge coefficients consistently identify supermarkets, laboratories, hospitals, data centres, restaurants, and recreation uses as high-intensity groups. Building age, floor area composition, location, and reporting year provide additional but weaker signals.

Q2 - How accurately can the models predict unseen buildings? Accuracy is useful but limited. The tuned Random Forest explains about 42.0% of test variation on the log scale and reduces raw MAE from 27.07 to 20.49 kBtu/sf. Large specialist-building errors remain, so predictions are screening estimates rather than substitutes for measured energy data.

Q3 - Which approach performs best? Tuned Random Forest on the full, leakage-safe encoded feature set performs best on the declared log- R^2 metric and MAE. Ridge is close and gives the best raw R^2 /RMSE plus clearer coefficient interpretation. Truncated SVD was not kept because the full feature representation preserved interpretability without a meaningful validation penalty.

8. Limitations and Next Steps

The data are annual self-reports and may contain reporting corrections, inconsistent use classifications, and changes in building operation. The model does not observe occupancy density, operating hours, weather at the property level, equipment condition, retrofit history, tenant behaviour, or detailed end uses. The 800-kBtu/sf ceiling makes the model more stable but means conclusions do not cover the most extreme records. Associations should not be interpreted causally.

With more time, the next steps are to add weather variables fitted by reporting year, use repeated-building history without leaking the test period, tune gradient-boosted trees, calculate confidence intervals through grouped resampling, and investigate separate models for major property types. External validation on another city's benchmarking data would test generalisability.

9. Contributions

Member Student ID Contribution

Tashrique Tanvir [23301389] Data audit, preprocessing, feature engineering, modelling, tuning, evaluation, visualisation, error analysis, and report preparation.

References

- City of Seattle. *Building Energy Benchmarking Data, 2015-Present*. <https://data.seattle.gov/d/teqw-tu6e>. Public domain.
- City of Seattle Office of Sustainability and Environment. *Energy Benchmarking Program*. <https://www.seattle.gov/environment/climate-change/buildings-and-energy/energy-benchmarking>.
- Libraries: pandas, NumPy, scikit-learn, matplotlib, and seaborn.
- **AI assistance disclosure:** ChatGPT (OpenAI) was used to help structure the analysis in line with the faculty template, check leakage safeguards, reproduce model-evaluation calculations, and draft this report from the dataset and supplied notebook figures. The student remains responsible for reviewing the code, values, interpretations, and final submission.

